8.5.2 Testing is no different

One of Akka's powerful abstractions is that you don't need to add anything additional to test your actors. You can emulate your application without having to start a cluster. The following listing shows how to create a test that proves that the master generates tick after tick and gets the expected number of words each time.

Listing 8.13 Testing the master and worker processing words

```
class WordsSampleSpec ... {

  "The words app" should {
    "send work from the master to the workers and back" in {
      //given
      val numberOfWorkers =
        system.settings.config
          .getInt("example.countwords.workers-per-node")

      for (i <- 0 to numberOfWorkers) {                    ❶
        spawn(Worker(), s"worker-$i")
      }

      val router = spawn {
        Routers
          .group(Worker.RegistrationKey)                   ❷
      }
      //when
      val probe = createTestProbe[Master.Event]
      val masterMonitored =
        Behaviors.monitor(probe.ref, Master(router))
      spawn(masterMonitored, "master0")                    ❸
      //then
      probe.expectMessage(Master.Tick)
      probe.expectMessage(
        Master.CountedWords(
          Map(
            "this" -> 1,
            "a" -> 2,
            "very" -> 1,
            "simulates" -> 1,
            "simple" -> 1,
            "stream" -> 2)))
      ...                                                  ❹
```